Demystifying AI

A Webinar from Pilot - UK AI Factory Antenna

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EPCC

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Pilot UK AI Factory Antenna
EuroHPC Joint Undertaking
Co-funded by the European Union
Department for Science, Innovation & Technology


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Preface

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AI involves…

- Hype
- Genuine potential for impact across nearly every domain / application area
- New ideas
- Old ideas
- Rapid change
- Uncertainty and differences of opinion
- Lots of data
- Lots of computational power
- Ethical questions and tradeoffs

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Ethical Considerations of AI

- Environmental
- Risk of bias
- Respect for intellectual property rights
- Consolidation/centralisation of power for Big Tech
- Implications for labour forces
- Risk of cognitive offloading
- Disinformation & misinformation
- Creation of abusive imagery
- Mass surveillance
- Applications in warfare

The rest of today's talk

- What is AI, and why is it now that it's taking off?
- AI vs Machine Learning vs Deep Learning
- An introduction to Machine Learning
- An introduction to Neural Networks
- A high-level overview of what happens behind the scenes when you type a prompt into ChatGPT (or similar tool) and press Return

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What is AI?

- Getting a computer to act as if it is intelligent

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What is AI?

- “Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making” -- wikipedia
- The term “Artificial Intelligence” is generally credited to **John McCarthy** in 1955: “Every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it.” And later, “Artificial intelligence is the science and engineering of making intelligent machines.”

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What is AI?

- “Artificial intelligence (AI) is technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy.” – Cole Stryker, Eda Kavlakoglu, IBM
- “Artificial Intelligence (AI) can be defined in many ways. However, within this guidance, we define it as an umbrella term for a range of algorithm-based technologies that solve complex tasks by carrying out functions that previously required human thinking.” – UK ICO
- “Artificial intelligence (AI) is a broad category of software that can recognize patterns, learn from data, and produce useful outputs”. – OpenAI

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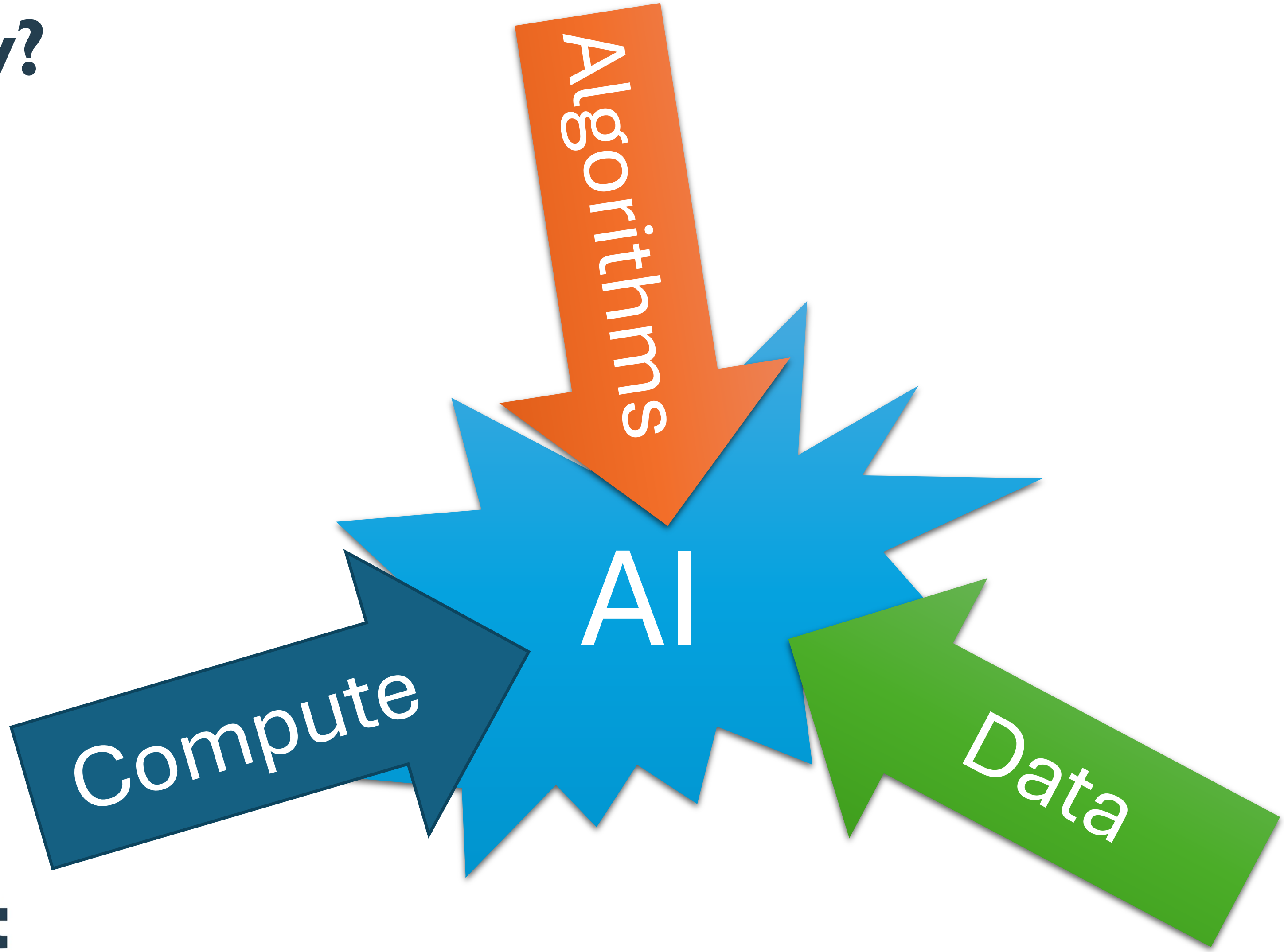
Why now?



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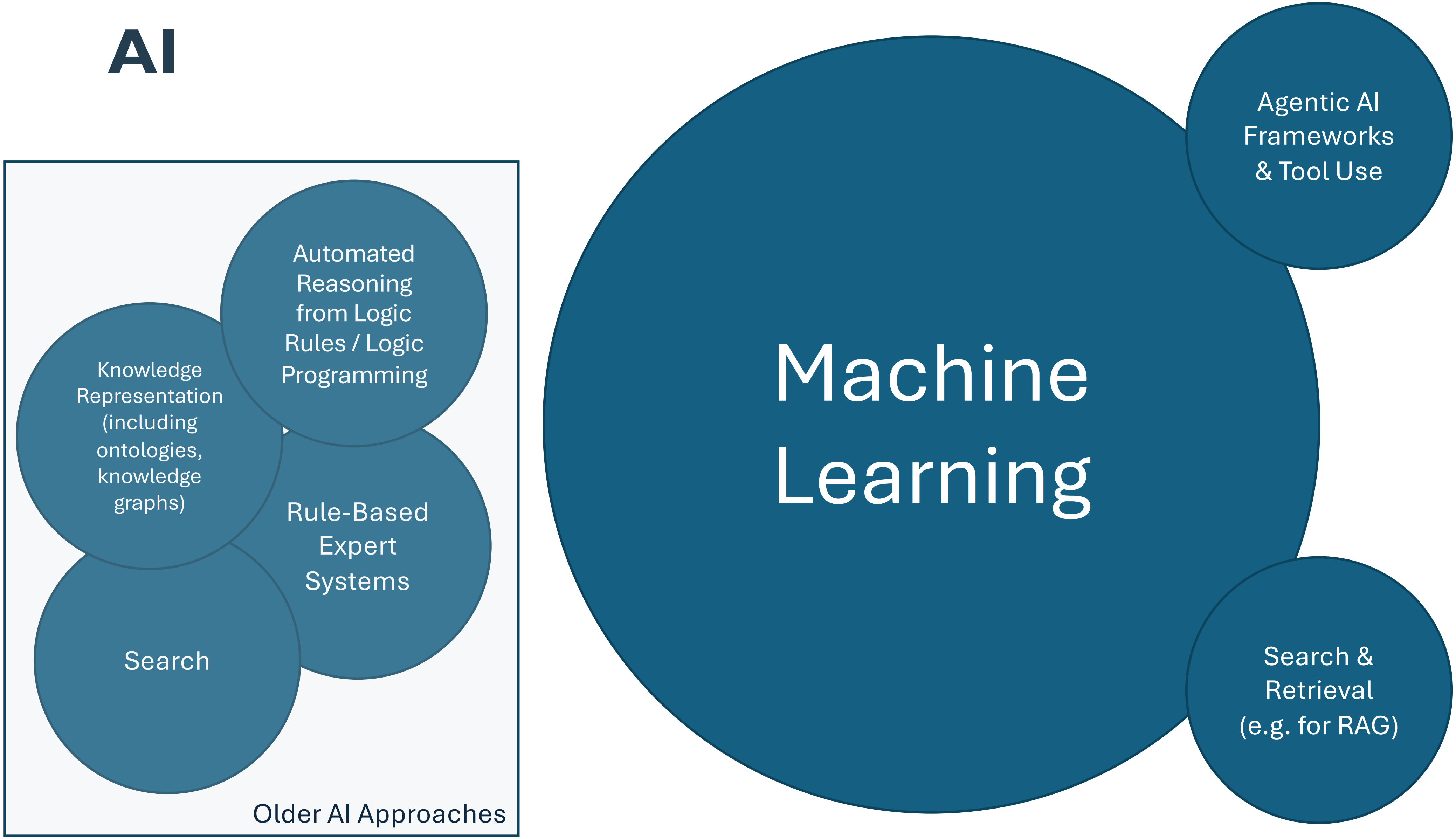
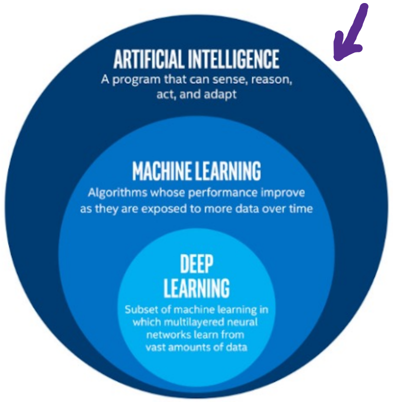
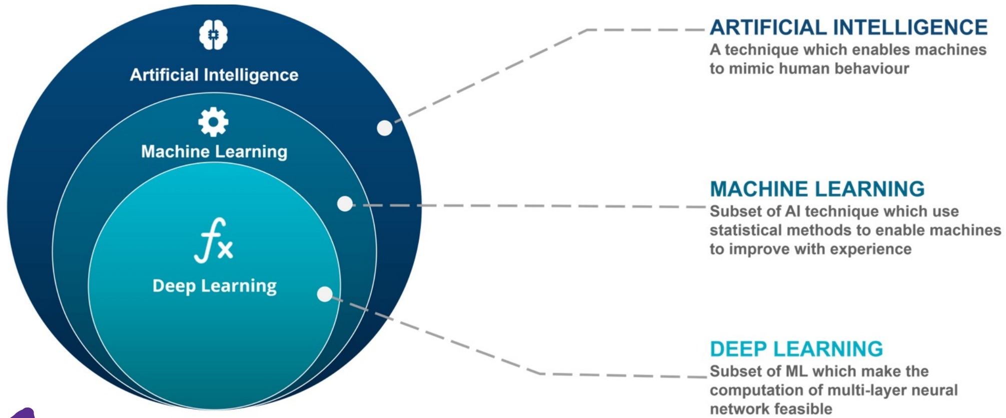

AI
Automated Reasoning from Logic Rules / Logic Programming
Knowledge Representation (including ontologies, knowledge graphs)
Rule-Based Expert Systems
Search
Older AI Approaches
Machine Learning
Agentic AI Frameworks & Tool Use
Search & Retrieval (e.g. for RAG)

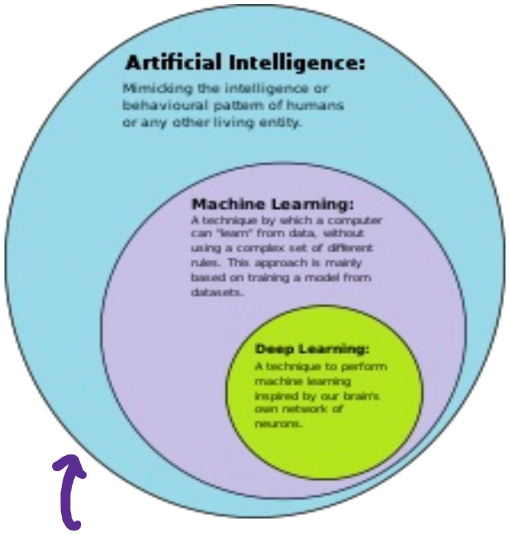
Machine Learning, Deep Learning & AI

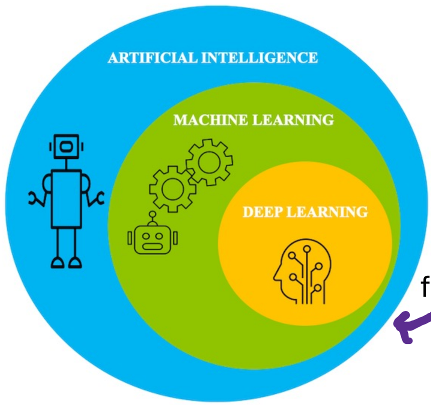
from https://builtin.com/machine-learning/what-is-deep-learning





from https://www.edureka.co/blog/ai-vs-machine-learning-vs-deep-learning/





from https://developer.ibm.com/articles/an-introduction-to-deep-learning/

from https://commons.wikimedia.org/wiki/File:AI-ML-DL.svg

What is Machine Learning?

- Getting a computer to do something, without explaining to the computer exactly how to do it
  - “Learn from these examples” – Supervised Learning
    - e.g., given many examples of pictures of cats and dogs with the label “cat” or “dog”, learn how to identify whether a new, unlabeled image is a cat or a dog
  - “Find patterns in this data” – Unsupervised Learning
    - e.g., given information about all your customers, can you group your customers into sets that are similar in some respect, perhaps so you could treat them as a group in a marketing campaign?
  - “Try to achieve this goal by learning from your experience” – Reinforcement Learning
    - e.g. get a robot to learn how to climb over a box by trying different movements of its legs to see which ones are most effective

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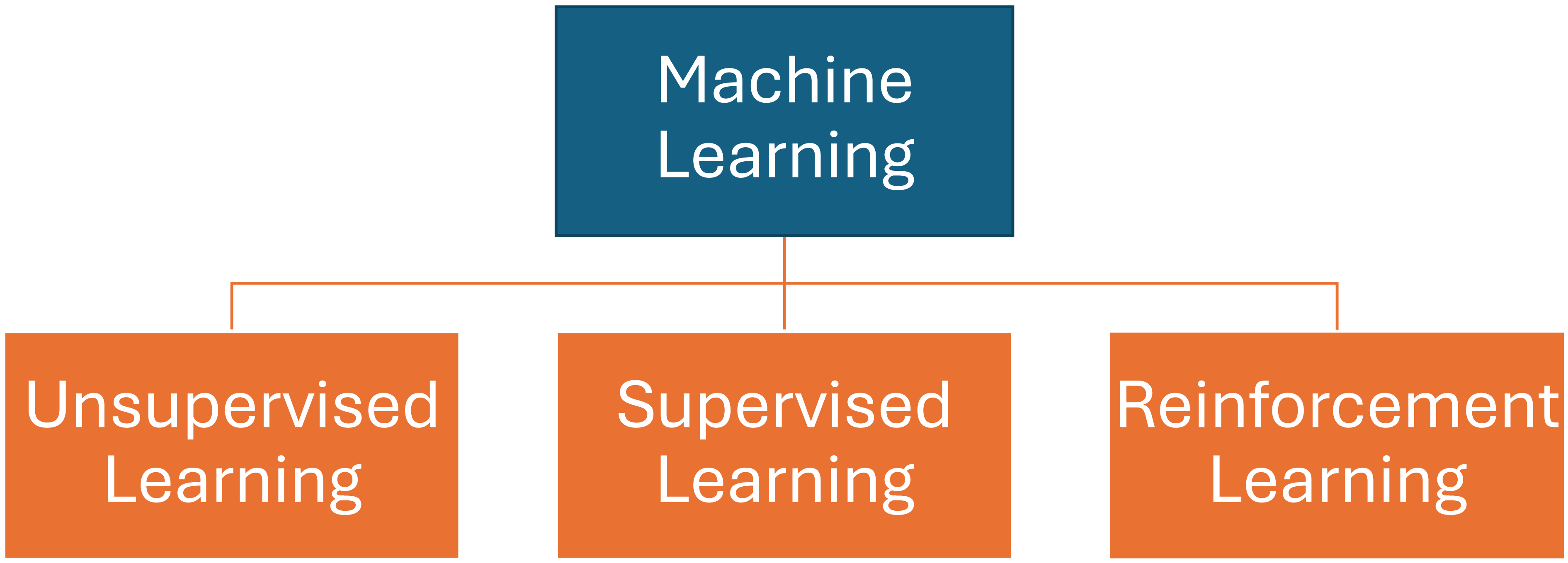

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Why is it so powerful?

- You can use (basically) the same algorithms for tasks in multiple domains, e.g.,
    - identifying genes that are similar to each other and identifying customers that are similar to each other
    - predicting which of your customers are likely to unsubscribe from your service and predicting the upcoming weather from current rain radar images
    - identifying tumours in medical images and identifying words spoken to a voice assistant
    - classifying images of cats and dogs, and classifying email as spam or not spam
    - teaching a robot to walk, and predicting the next word you will type in a text message
    - responding to a prompt in ChatGPT and creating an image from a text prompt

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Some other definitions of machine learning

- “the field of study that gives computers the ability to learn without explicitly being programmed.” – Arthur Samuel, IBM (~1959)
  - This is possibly a paraphrase, but it is widely accepted that the term was coined by Arthur Samuel
- “Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can effectively generalize and thus perform tasks without explicit instructions” – Wikipedia (accessed November, 2023)
- “Machine learning is a branch of artificial intelligence (AI) and computer science which focuses on the use of data and algorithms to imitate the way that humans learn, gradually improving its accuracy.” – IBM
- “it's the science of getting computers to learn without being explicitly programmed” – Andrew Ng, Stanford/Coursera/…

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What do the machines learn?

- Central to Machine Learning (ML) is the idea of a **model**
- A model is learned from data using a **machine learning algorithm** and the model can then be used to make predictions/classifications from new, previously unseen data (in the case of supervised learning) or to cluster/group the data (in the case of unsupervised learning) or to decide on a course of action in an observed environment (reinforcement learning)
- Generally speaking, with ML, machines do *not* learn facts or build an understanding the way humans do; they learn models/patterns
  - No reason why this isn't possible in future, but that's not how ML currently works, even in ChatGPT!

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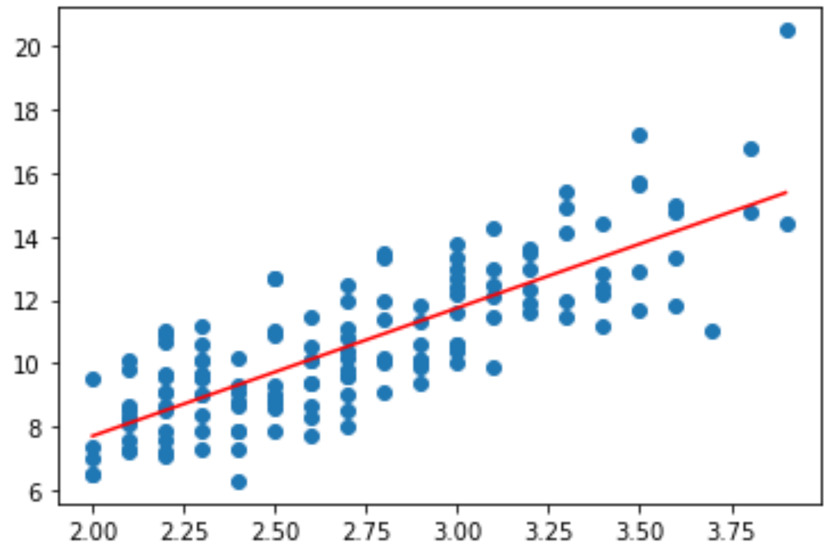
Linear Regression as an example of ML

- Given a new value of data point with known *x* but unknown *y* can you predict *y*?
- The **model** here is the red line
- The model is **learned** using a process which finds the best fit for the existing datapoints: the "**training data**"
- We sometimes say we **fit** the model to the training data, or that we **train** the model using the training data
- Key point: In ML, like here, we often decide on the shape of the model (here a straight line) but we let the machine learning algorithm find what the *best* straight line is...



Training

Training Set (Model)

5.1, 1.4, setosa
5.7, 1.3, setosa
…
6.3, 6.0, virginica

Model Architecture

e.g. $\beta_0 + \beta_1 x_1 = 0$

Definition of similarity

e.g. Euclidian distance

Hyperparameters

e.g. k=3

Model

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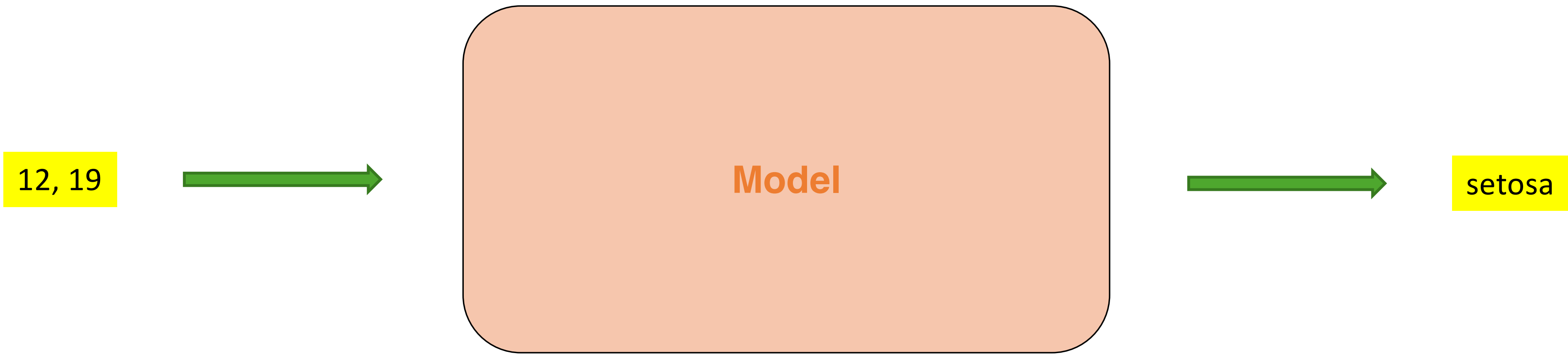
Inference



Some important ideas

- Models are only as good as the data they're trained on
  - Possibility for bias, changing underlying conditions
  - Rare events can be harder to model (less training data)
- Machine learning models are often opaque:
  - It can be difficult to "understand" why they make a certain prediction
- There are different ways of deciding if a model is "good"
  - Would you prefer your AI poisonous mushroom predictor to be 98% accurate, or would you prefer that it was only accurate 95% of the time but erred on the "safe" side when unsure?
- Training models is often computationally expensive
  - Although modern computing resources make this more accessible than ever
  - Training is usually much more resource intensive than prediction/inference

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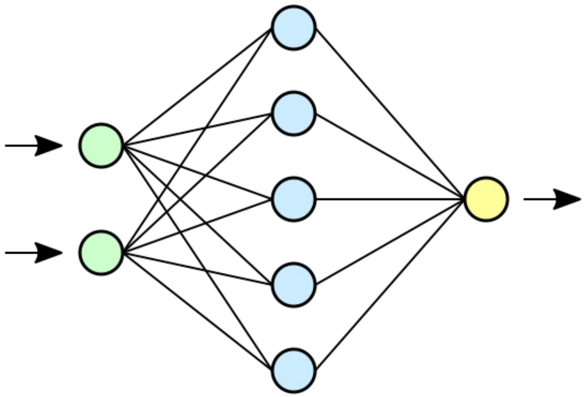
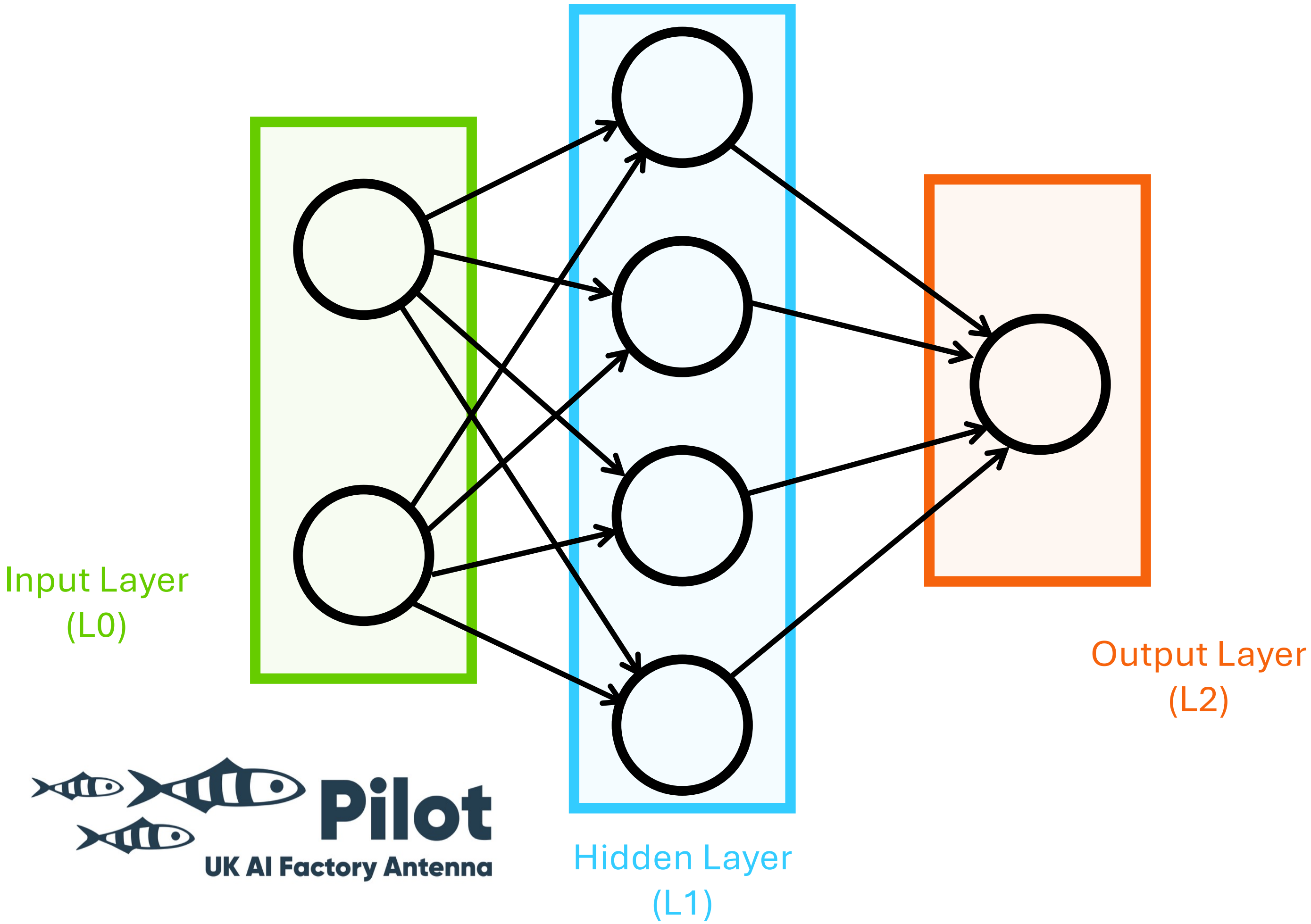
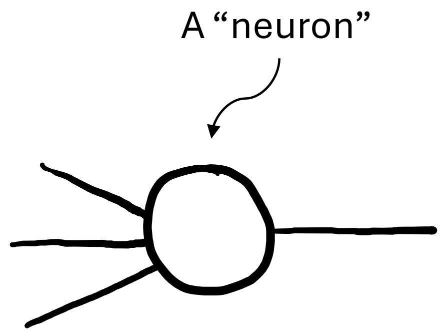
What is a Neural Network?

- "A neural network is a network or circuit of biological neurons, or, in a modern sense, an artificial neural network, composed of artificial neurons or nodes." – Wikipedia
- "Neural networks, also known as artificial neural networks (ANNs) or simulated neural networks (SNNs), are a subset of machine learning and are at the heart of deep learning algorithms. Their name and structure are inspired by the human brain, mimicking the way that biological neurons signal to one another." – IBM Cloud Education
- "A family of parametric, non-linear and hierarchical representation learning functions, which are massively optimized with stochastic gradient descent to encode domain knowledge, i.e. domain invariances, stationarity." - Efstratios Gavves



Components of a Neural Network





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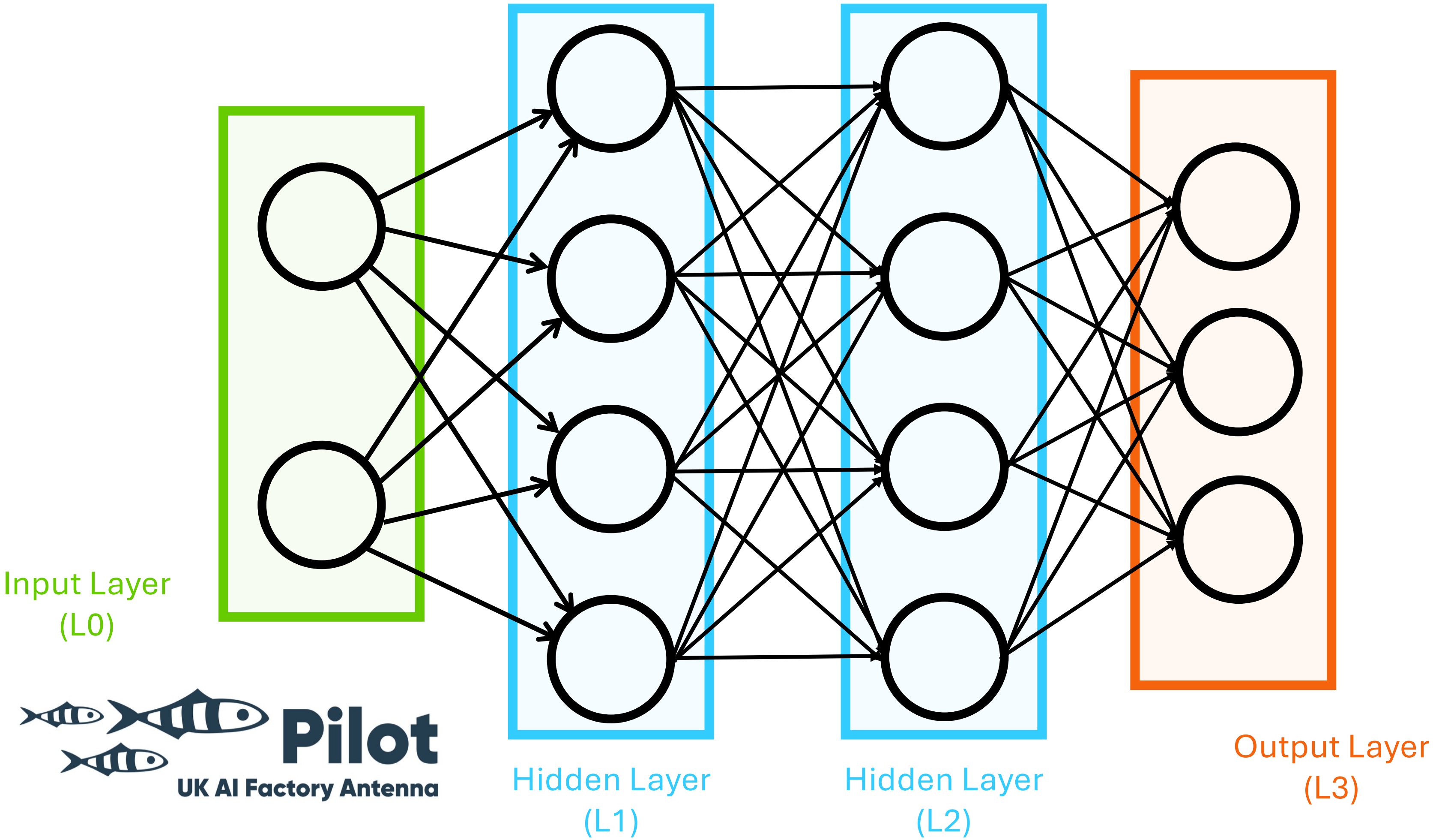

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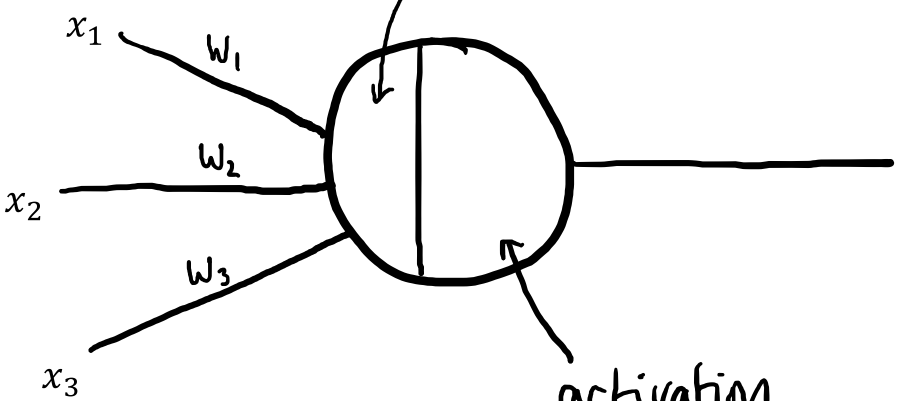
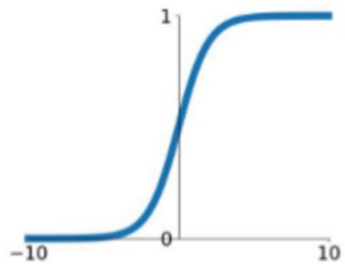
A “neuron”

$$z = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$





activation function, e.g. $a = \sigma(z) = \frac{1}{1+e^{-z}}$

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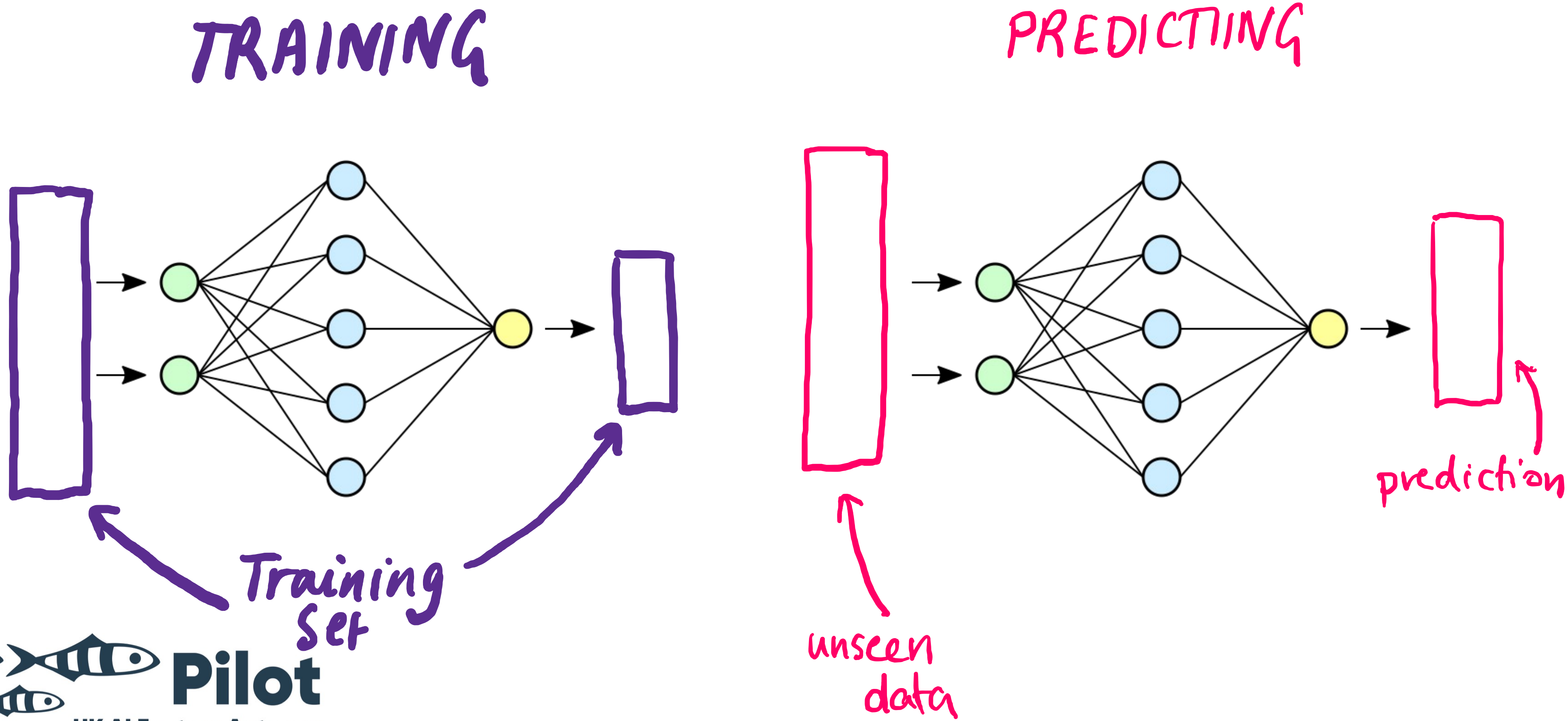
Training & Inference with a Neural Network



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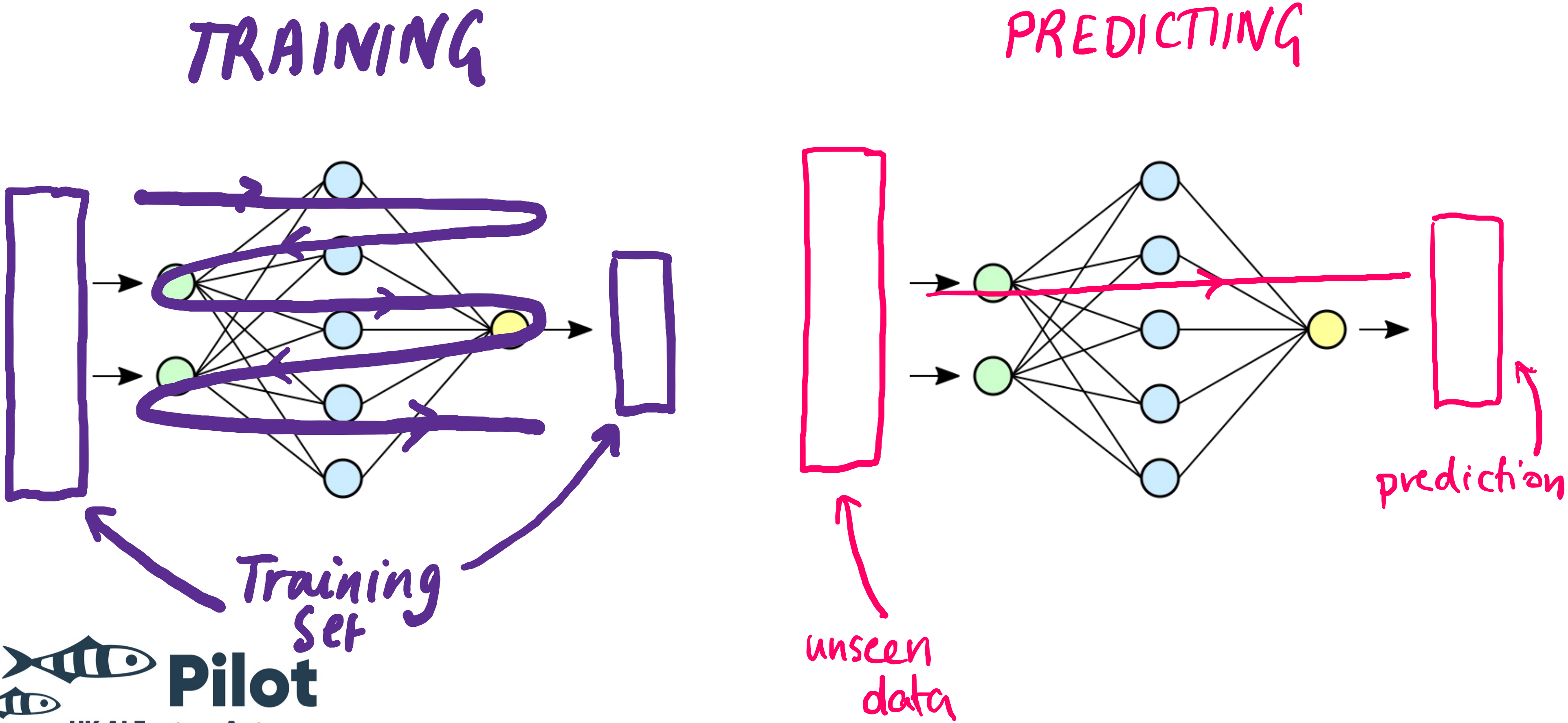

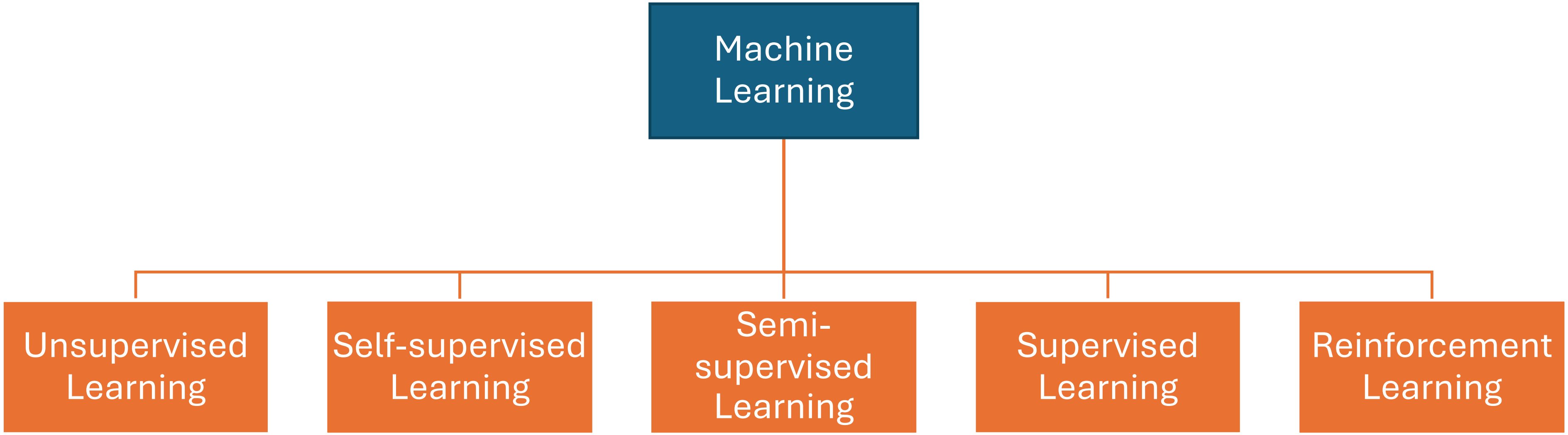

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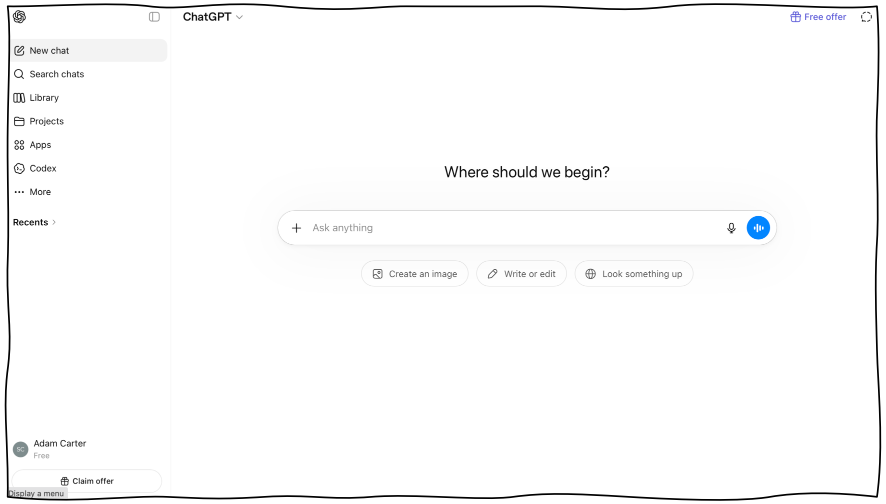

What happens when you use an AI assistant?

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The Basic Idea
(An oversimplification to highlight key ideas)

- Tokenisation & Embedding Look Up
  - "your prompt is broken up into tokens" -> "your" " prompt" " is" " broken" " up" " into" " tokens" -> [4531, 10137, 374, 11234, 709, 1139, 11460]
  - Embedding look up: 4531 → [0.23, -1.17, ..., 0.88]

A long vector of numbers (*GPT-2 small* uses a length of 768)

- The vectors are "stacked" into a stack of vectors
- This stack of vectors is passed into the input layer of a neural network for a single "feed-forward" / inference step
- The neural network ultimately outputs another high-dimensional vector, this time with an entry associated with each possible output word
- The word (token) with the highest value is chosen as the next word
- If the output token is a special <END OF STREAM> token, the process stops
- Otherwise, this word is added to the prompt, and the process is repeated

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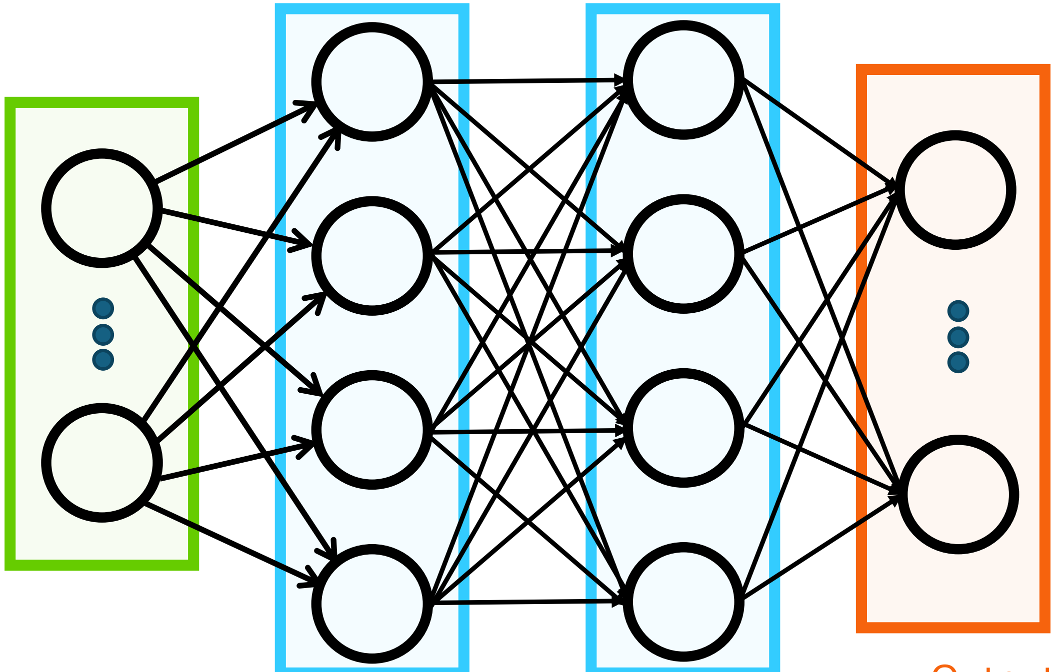
Want to see the full list of 50,257 tokens used by GPT-2? See https://huggingface.co/openai-community/gpt2/raw/main/vocab.json

The Basic Idea
(An oversimplification to highlight key ideas)

A prompt with 100 tokens using GPT-2 small's 768-length embedding $100 \times 768 = \mathbf{76,800}$ neurons in the **input layer**



With a dictionary of size 50,257, the **output layer** would have **50,257** neurons

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A few additional details

0. The prompt is not just what you type in:
   - It is combined with:
     - System instructions (invisible to the user)
     - *Every previous prompt and response in the "conversation"*
       - This **context** grows as the conversation goes on making things more computationally expensive/slow as the conversation grows

1. Tokenisation & Embedding Look Up
   1. "your prompt is broken up into tokens" -> "your" " prompt" " is" " broken" " up" " into" " tokens" -> [4531, 10137, 374, 11234, 709, 1139, 11460]
   2. Embedding look up: 4531 → [0.23, -1.17, ..., 0.88]

1a. The embedding vector added to another vector that encodes position

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Some more additional details

- Sometimes the prompt is passed through a preprocessing pipeline (either with pattern matching or being passed to a smaller routing model) to help with:
  - Safety / moderation / policy enforcement
  - Selection of models (e.g. to decide to use a more complex model or whether a simple model would be sufficient to answer the question)
  - Tool calling (e.g., do I need to perform a web search to address a question in the prompt?)

- Current products have started to adopt strategies used in agentic AI systems:
  - They can loop several times (with hidden outputs) to perform multiple "reasoning" steps, calling tools in intermediate stages (e.g., to run python code)

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Demystifying AI Webinar Follow Up

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